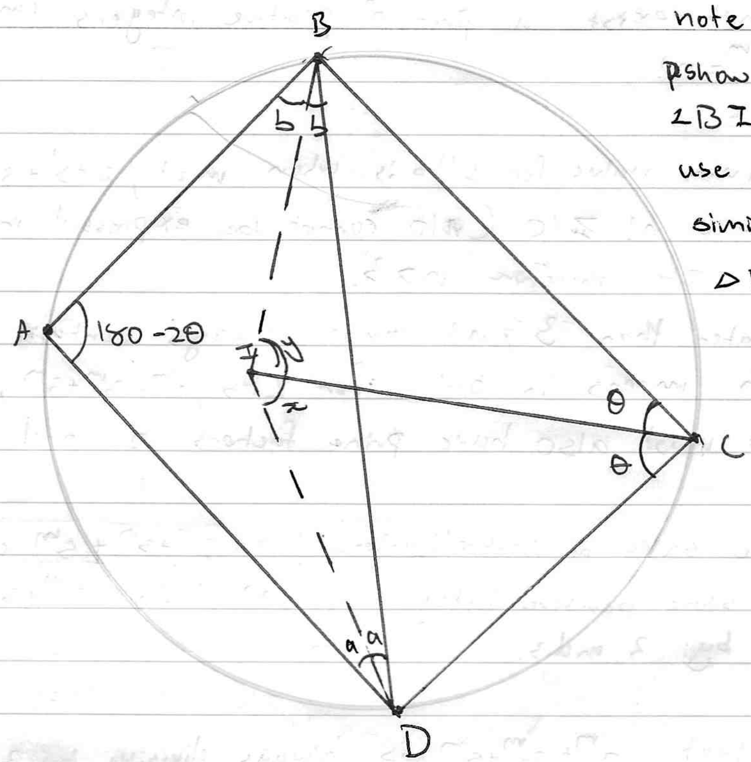
Solution for problem 2



note that we want to show that $\angle IDC$ equal $\angle BIC$ so that we can use angle - angle similarity to show that $\triangle DIC \sim \triangle IBC$

let $\angle BCD = 2\theta$, since CI bisects $\angle BCD$ $\angle BCI = \angle DCI = \theta$

let $\angle ADI$ be a, since I is the incentre of $\triangle ABD$, then $\angle IDB = \angle ADI = a$

let $\angle ABI$ be b, since I is the incentre of $\triangle ABD$, then $\angle DBI = \angle ABI = b$

~~Angles~~ the sum of opposite angles in a cyclic quadrilateral is 180°, therefore $\angle BAD + \angle BCD = 180^\circ$ and thus $\angle BAD = 180 - 2\theta$.

let ~~x be the~~ $\angle DIC$ be x and let $\angle CIB$ be y.

Angles in triangles sums up to 180°, thus, $180 - 2\theta + 2b + 2a = 180$

$\Rightarrow -2\theta + 2b + 2a = 0 \Rightarrow 2\theta = 2(a+b) \Rightarrow \theta = a+b$ $(\triangle ABD)$ ③

and $a + x + y + b = 180^\circ$ $(\triangle DIB)$ ①

and $\theta + x + \angle IDC = 180^\circ$ $(\triangle IDC)$ ②

$\Rightarrow a + x + y + b = \theta + x + \angle IDC$ ① = ② (They both equal 180°)

$\Rightarrow a + x + y + b = a + b + x + \angle IDC$ ③ $(\theta = a + b)$

$\Rightarrow \angle IDC = y$.

Since $\angle IDC = \angle BIC = y$ and $\angle ICB = \angle DCI = \theta$

then by angle-angle similarity of a triangle, $\triangle IDC \sim \triangle BIC$

or $\triangle BCI \sim \triangle ICD$.